

Question_8:

Code:

```
1  /*
2  Question_8:
3  Find the minimum extra petrol required to cover all sub-tracks.
4  If no extra petrol is needed, print -1.
5  */
6
7  #include <iostream>
8  using namespace std;
9
10 void sort(int a[], int n) {
11     for (int i = 0; i < n - 1; i++)
12         for (int j = 0; j < n - i - 1; j++)
13             if (a[j] > a[j + 1])
14                 swap(a[j], a[j + 1]);
15 }
16 int main() {
17     int T;
18     cin >> T;
19
20     while (T--) {
21         int n, k;
22         cin >> n >> k;
23
24         int a[n];
25         for (int i = 0; i < n; i++)
26             cin >> a[i];
27         sort(a, n);
28
29         if (a[n - 1] <= k)
30             cout << -1 << endl;
31         else
32             cout << a[n - 1] - k << endl;
33     }
34     return 0;
35 }
```

Output:

Test Case_1:

```
2
5 6
2 5 4 5 2
-1
5 3
1 6 3 5 2
3
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
2
5 6
12 15 14 15 12
9
5 3
11 61 31 51 21
58
PS C:\Users\Prabin Yadav\
```